

Vanuatu NBSAP GIS Portal — Methodology Report

Geospatial Calculation, Analysis & Reporting Methodologies

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1. Introduction

The Vanuatu NBSAP GIS Portal is a geospatial analysis and reporting platform that tracks progress toward nine Global Biodiversity Framework (GBF) targets. It implements dissolution-based area calculation methodology following UNEP-WCMC standards, ensuring each point on Earth's surface is counted only once toward coverage targets, preventing double-counting of overlapping designations.

1.1 National Baselines

All percentage-based metrics are calculated against Vanuatu's national baselines:

Baseline	Area (ha)	Area (km ²)
Terrestrial (land)	1,219,000	12,190
Marine (sea)	66,300,000	663,000

1.2 Coordinate Reference System

All geospatial data is standardised to **EPSG:4326 (WGS 84)** using geodesic area calculations via Turf.js, which accounts for Earth's curvature.

1.3 Province Framework

Analysis is disaggregated by Vanuatu's six provinces:

2. Core Area Calculation Methodology

2.1 Geodesic Area Computation

All area calculations use **geodesic measurement** on WGS 84 coordinates via `turf.area()`, which computes the area of a polygon on an ellipsoidal Earth model. Results are converted from square metres to hectares:

$$\text{Area (ha)} = \text{turf.area}(\text{polygon}) / 10,000$$

This method produces accurate results for Vanuatu's equatorial-to-subtropical latitude range without requiring projection to a local coordinate system.

2.2 Gross Area vs Net Area

The portal distinguishes between two area measures:

Measure	Definition	Method
Gross Area	Sum of individual feature areas	Simple addition of <code>area_ha</code> per feature
Net Area	Actual geographic coverage (overlap removed)	Polygon dissolution (union) then area recomputation

Gross area treats each designation independently — if two protected areas overlap by 100 ha, both contribute their full area. **Net area** merges overlapping geometries so that shared land is counted only once.

2.3 Polygon Dissolution (Union) — UNEP-WCMC Methodology

Dissolution follows the UNEP-WCMC / GBF methodology for calculating effective conservation coverage. Overlapping features are unioned into a single merged geometry before area is recomputed.

Algorithm

1. **Filter:** Extract Polygon and MultiPolygon features (points and lines excluded from area calculations)

2. **Size check:**
3. 0 polygons → return null (no area)
4. 1 polygon → return as-is (no overlap possible)
5. >500 polygons → skip dissolution, fall back to gross area (performance safeguard)
6. **Chunked tree-merge** (for 100–500 polygons):
7. Split into chunks of 100 features
8. Dissolve each chunk using `turf.union()`
9. Recursively merge chunk results
10. **Direct dissolution** (<100 polygons):
11. Batch union via `turf.union(featureCollection)`
12. Fallback to iterative pairwise union if batch fails
13. Invalid geometries are logged and skipped

Overlap Indicator

When net area differs from gross area by more than 1 ha, the overlap percentage is reported:

$$\text{Overlap \%} = (1 - \text{Net Area} / \text{Gross Area}) \times 100$$

This quantifies the degree of spatial overlap between designations.

2.4 Reference Layers

Layers marked as **reference** (`isReference: true`) are displayed on maps for context but are **excluded from all area calculations, totals, and statistics**. Examples include administrative boundaries and historical comparison datasets.

3. Spatial Join — Province Assignment

Features are assigned to provinces using a point-in-polygon spatial join:

1. **Centroid extraction:** Each feature's effective centroid is calculated:
2. Point features → used directly
3. Polygon features → geometric centroid
4. MultiPolygon features → centroid of the largest polygon part
5. **Bounding box pre-filter:** Province bounding boxes are pre-computed; features outside a province's bbox are rejected before the expensive point-in-polygon test (~80% rejection rate)

6. **Point-in-polygon test:** `turf.booleanPointInPolygon()` determines which province the centroid falls within
7. **Assignment:** The first matching province name is assigned to `feature.properties.province`

Features that fall outside all province boundaries retain their original province value or are marked "Unassigned" and excluded from provincial breakdowns.

4. Metrics Computation by Target

4.1 Target 3 — 30×30 Conservation (Protected Areas & OECMs)

GBF Goal: Protect 30% of terrestrial and 30% of marine areas by 2030.

Categories: CCA (Community Conserved Areas), MPA (Marine Protected Areas), LMMA (Locally Managed Marine Areas), PA (Protected Areas — WDPA), OECM (Other Effective Conservation Measures)

Methodology:

1. **Filter:** Only layers with `countsToward30x30 = true` and `targets` including T3 are included
2. **Realm separation:** Features are split into terrestrial and marine groups based on `realm` property (or category default realm)
3. **Gross computation:** Sum of all `area_ha` values per realm
4. **Dissolution:** Terrestrial and marine feature sets are dissolved separately using polygon union
5. **Net computation:** Area recomputed from dissolved geometries
6. **Percentage:** $\text{Coverage \%} = (\text{Net Area} / \text{National Baseline}) \times 100$
7. **Gap analysis:** $\text{Remaining \%} = \max(0, 30 - \text{Coverage \%})$; $\text{Remaining ha} = \text{Remaining \%} \times \text{Baseline} / 100$

Province breakdown: Features filtered per province, dissolved per realm within each province, producing per-province terrestrial and marine coverage figures.

Output metrics: - Net terrestrial ha and % of national land - Net marine ha and % of national sea - Remaining % and ha to reach 30% for each realm - Province breakdown (terrestrial, marine, total per province) - Category breakdown (area per designation type)

4.2 Target 1 — Biodiversity-Inclusive Spatial Planning

GBF Goal: 100% of land and sea areas covered by biodiversity-inclusive spatial plans.

Categories: SPATIAL_PLAN (Provincial & Municipal Physical Plans), KBA (Key Biodiversity Areas), CCA (Community Conserved Areas)

Methodology:

1. **Filter:** Layers with `targets` including T1
2. **Dissolution:** All features dissolved together; also dissolved by category and by realm
3. **Coverage:** Net area as % of national terrestrial and marine baselines
4. **Category breakdown:** Net area per category (SPATIAL_PLAN, KBA, CCA)
5. **Province breakdown:** Per-province dissolved area

Key metrics: Total spatial planning coverage (ha), % of national area, number of provinces with planning data, category contributions.

4.3 Target 2 — Degraded Area Mapping & Ecosystem Restoration

GBF Goal: Map all degraded areas; restore 30% of degraded ecosystems.

Categories: DEGRADED (Degraded Terrestrial and Marine/Coastal Areas), RESTORATION (Active & Planned Restoration Sites)

Methodology:

1. **Filter:** Layers with `targets` including T2
2. **Category split:** DEGRADED and RESTORATION features computed separately
3. **Dissolution:** Each category dissolved independently to compute net degraded area and net restoration area
4. **Overlap:** Overlap between degraded and restoration layers quantified
5. **Province breakdown:** Per-province degradation and restoration extent

Key metrics: Total degraded area (ha), total restoration area (ha), overlap between designations, province-level breakdown.

4.4 Target 4 — Species & Biodiversity Distribution

GBF Goal: Map distribution of significant endemic and threatened species.

Priority species tracked:

Species	Scientific Name	Taxa
Vanuatu Megapode	Megapodius layardi	Bird
Vanuatu Mountain Starling	Aplonis santovestris	Bird (critically endangered)
Vanuatu Streaked Fantail	Rhipidura spilodera	Bird
Vanuatu Kingfisher	Todiramphus farquhari	Bird
Vanuatu Flying Fox	Pteropus anetianus	Mammal
Plerandra vanuatuensis	Plerandra vanuatuensis	Plant (endemic)

Additional categories: KBA, SPECIES_DIST (general species distribution)

Methodology:

1. **Filter:** Layers with targets including T4
2. **Per-species metrics:** Each species category (MEGAPODE, STARLING, etc.) dissolved separately
3. **Distribution mapping:** Both point (observation records) and polygon (range/habitat) data supported
4. **Coverage:** Net area per species, total distribution area
5. **Completeness:** Species mapped vs expected species (6 priority + KBA + general)

Key metrics: Number of species mapped, distribution area per species, total observation records, province distribution, data completeness (% of expected species).

Note: Only polygon features contribute to area calculations. Point observations contribute to record counts and distribution mapping but not area totals.

4.5 Target 6 — Invasive Alien Species (IAS)

GBF Goal: Map coverage and distribution of priority invasive alien species for management planning.

Priority IAS tracked:

Species	Scientific Name	Threat Description
Merremia peltata (Big Leaf)	Merremia peltata	Invasive vine smothering canopy trees, preventing forest regeneration
Crown of Thorns Starfish	Acanthaster planci	Outbreaks causing devastating coral mortality across reef systems
Mile a Minute Vine	Mikania micrantha	Fast-growing vine forming dense mats over native vegetation
Devil Fig	Solanum torvum	Woody shrub colonising disturbed land, displacing native undergrowth

Additional category: INVASIVE (other IAS — Fire Ants, African Snail, Sako, Coconut Beetle)

Methodology:

1. **Filter:** Layers with `targets` including T6
2. **Per-species analysis:** Each IAS category dissolved separately
3. **Total IAS extent:** All IAS features dissolved together for net coverage
4. **Province distribution:** Per-province IAS presence and extent
5. **Type breakdown:** Detection methods, confidence levels where available

Key metrics: Total IAS-affected area (ha), per-species area, provinces affected, detection records, % of national terrestrial area affected.

4.6 Target 10 — Agriculture & Land Cover Change

GBF Goal: Map land cover change for agriculture, livestock, fisheries, and forestry sectors.

Category: LAND_COVER (Land Cover / Land Use)

Methodology:

1. **Filter:** Layers with `targets` including T10
2. **Type disaggregation:** Features grouped by `type` property (land use classification):
3. Examples: Prime Agricultural Land, Non Agricultural Land, Other Agricultural Land, Marginal Agricultural Land
4. Dynamic colour assignment for sub-types not in the predefined palette

5. **Per-type dissolution:** Each land use type dissolved separately for net area
6. **Province breakdown:** Per-province land cover composition
7. **Change detection context:** Type breakdown enables temporal comparison when multiple years uploaded

Key metrics: Total area by land use type, number of distinct land use classes, per-province composition, records per type.

Performance note: T10 datasets can contain thousands of features. When feature counts exceed 500, dissolution is skipped and gross area is used as the best available estimate to maintain portal responsiveness.

4.7 Target 7 — Pesticide & Herbicide Mapping

GBF Goal: Map areas of chemical use in farming to assess biodiversity risk.

Category: PESTICIDE

Methodology: Standard target metrics — dissolution of all pesticide-use areas, breakdown by chemical type and province.

4.8 Target 8 — Coastal Eutrophication

GBF Goal: Monitor nutrient-impacted marine zones threatening reef health.

Category: EUTROPHICATION (marine realm)

Methodology: Standard target metrics — dissolution of eutrophication zones, breakdown by severity level and province.

4.9 Target 12 — Blue & Green Spaces

GBF Goal: Map urban parks, botanical gardens, and green infrastructure.

Category: GREEN_SPACE

Methodology: Standard target metrics — dissolution of green space polygons, breakdown by space type and province.

5. Analysis & Reporting Framework

5.1 Data Completeness Assessment

For each target, the portal calculates **data completeness** as the percentage of expected data layers that have been uploaded:

$$\text{Data Completeness \%} = (1 - \text{Missing Layers} / \text{Expected Layers}) \times 100$$

Expected layers are defined per target in the system configuration.

5.2 Assessment Status

Each target (and each province within a target) receives a qualitative status:

Status	Criteria	Colour
No Data	No data layers uploaded	Red
Minimal	Data completeness < 50% and fewer than 2 layers	Red
Moderate	Data completeness ≥ 50% or 2+ layers uploaded	Orange
Comprehensive	Data completeness ≥ 80% and geometry quality ≥ 90%	Green

5.3 Key Findings (Auto-Generated)

The analysis engine generates quantitative findings based on the data:

1. **Coverage summary:** Total net area and % of national baselines
2. **Overlap detection:** Degree of spatial overlap between designations (significant if >5%)
3. **Province distribution:** Province with highest/lowest coverage
4. **Data gaps:** Provinces with no data for the target
5. **Target progress (T3):** Gap to 30% target in % and hectares
6. **Category concentration:** Whether coverage is dominated by a single designation type (>60%)
7. **Geometry types:** Mix of polygon vs point data and implications for area calculations
8. **Reference data:** Count of reference layers shown for context

5.4 Target-Specific Insights

Each target receives qualitative insights tailored to its GBF context:

- **T1**: Spatial planning coverage relative to mainstreaming biodiversity
- **T2**: Degradation baseline adequacy and restoration progress
- **T3**: 30×30 progress emphasising CCAs and LMMAs as culturally appropriate pathways for Vanuatu
- **T4**: Species coverage completeness and island-specific endemism
- **T6**: Per-species ecological threat descriptions and management prioritisation
- **T10**: Land cover type diversity and forest-to-agriculture conversion focus
- **T12**: Urban green infrastructure and biodiversity corridors

5.5 Province-Level Analysis

Each province receives individual analysis including:

- Province's contribution to the national total (ha and %)
- Per-province category breakdown
- Per-province type/species breakdown
- Province-specific data completeness
- Province share vs national total comparison
- Targeted findings and insights for the province

6. Map Production Methodology

6.1 Symbolology

Maps use consistent symbolology driven by a central configuration:

- Each category has a predefined fill colour and stroke colour
- LAND_COVER sub-types receive dynamically assigned colours from a 20-colour high-contrast palette using hash-based stable assignment
- Reference layers are rendered with dashed outlines and low fill opacity
- Province boundaries shown as context with permanent labels

6.2 Print Map Production

Print maps follow a standardised cartographic template:

1. **Title block:** Target name, province (if filtered), NBSAP branding, Vanuatu flag, date, CRS
2. **Map:** Leaflet-rendered map with CartoDB Light basemap, dissolved feature boundaries per symbology group, province boundaries, scale bar, north arrow
3. **Legend:** Category colours with labels, including LAND_COVER sub-type expansion
4. **Metrics row:** Terrestrial/marine net area, % national baselines, record count
5. **Category breakdown table:** Per-category area and record counts
6. **Analysis page:** Quantitative metrics, qualitative findings, insights

6.3 Dissolution on Maps

For cartographic output, features are dissolved per symbology group (category, or sub-type for LAND_COVER) following the same UNEP-WCMC union methodology used for metrics. For very large datasets (>500 features), raw features are rendered with a feature cap to maintain browser performance.

7. Data Pipeline & Quality Assurance

7.1 Upload Processing Pipeline

Uploaded datasets pass through a standardised pipeline:

1. **Validation:** File format check (GeoJSON, Shapefile, CSV with coordinates)
2. **Reprojection:** Transformation to EPSG:4326 (WGS 84) if necessary
3. **Geometry repair:** Automatic fix of self-intersections and ring-winding errors
4. **Field mapping:** Source field names mapped to canonical schema using category-specific lookup tables
5. **Province assignment:** Spatial join against province boundaries
6. **Area computation:** Geodesic area calculated for each polygon feature

7.2 Geometry Quality Tracking

The pipeline tracks three geometry quality indicators per layer:

Indicator	Description
Valid geometries	Features with geometries that passed validation
Fixed geometries	Features with geometries that were automatically repaired
Dropped geometries	Features with irreparably invalid geometries (excluded from analysis)

Geometry Quality % = (Total Features - Dropped) / Total Features × 100

7.3 Field Mapping

The system supports flexible field name mapping to accommodate diverse data sources. Each category defines acceptable input field names that are automatically mapped to canonical properties (`name` , `type` , `realm` , `province` , `year` , `status` , `source` , `notes`).

8. Summary — Methodology by Target

Target	Dissolution	Realm Split	Key Metric	Threshold
T1	By category & realm	Yes	Coverage % of national area	100% spatial plan coverage
T2	By category	Optional	Degraded + Restoration ha	Map all degraded areas
T3	By realm, province, category	Yes	% of 30% target achieved	30% terrestrial & marine
T4	By species category	No (terrestrial)	Species mapped / expected	All priority species mapped
T6	By IAS species	Optional	Total IAS extent (ha)	All priority IAS mapped
T7	Standard	No (terrestrial)	Pesticide use area (ha)	—
T8	Standard	No (marine)	Eutrophication zone (ha)	—
T10	By land use type	No (terrestrial)	Land cover composition	—
T12	Standard	No (terrestrial)	Green space area (ha)	—

9. Standards & References

- **UNEP-WCMC**: Methodology for calculating protected area coverage — polygon dissolution to prevent double-counting
- **Global Biodiversity Framework (GBF)**: Kunming-Montreal targets and indicators
- **WDPA**: World Database on Protected Areas — PA and OECM standards
- **Turf.js**: Geodesic spatial analysis library for area computation and polygon union

- **EPSG:4326 (WGS 84)**: Standard geographic coordinate reference system
 - **BirdLife International**: KBA Partnership methodology for Key Biodiversity Areas
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